

4x4 Matrix Keypad - ESP32 Interfacing Notes

Overview

A 4x4 matrix keypad is an input device with 16 keys arranged in 4 rows and 4 columns. It uses only 8 GPIO pins and works on row-column scanning technique.

Structure

The keypad is arranged as a matrix. Each key connects one row and one column. Example: Key '5' connects R2 and C2.

Working Principle

Rows are configured as OUTPUT and columns as INPUT with pull-up resistors. When a key is pressed, the corresponding row and column connect and a LOW signal is detected.

Scanning Method

The microcontroller sets one row LOW at a time and reads all columns. This process repeats continuously to detect key presses.

ESP32 Interfacing

Rows: GPIO14, GPIO27, GPIO26, GPIO25. Columns: GPIO33, GPIO32, GPIO22, GPIO23. Avoid using GPIO34 and GPIO35 as outputs because they are input-only pins.

Applications

Used in password systems, door locks, calculators, ATM machines, and embedded menu navigation systems.

Key Points

Use pull-up resistors for columns, implement debounce logic, and ensure correct row-column mapping for accurate detection.